



School of Engineering
Diploma in AI & Data Engineering

EGDT305

Big Data Processing Platforms &
Applications

Pipeline Planning

AY2026S1

Selected scenario:

Fraud Detection - car insurance claims.

Potential data sources:

- Claims & policy/underwriting system (internal, structured):

claim_id, policy_number, claim amounts, fraud_reported, bind_date, state, csl, premium, deductible, zip, age.

- Police incident reports (external, semi-structured):

police_report_available, claimant's account and ties to the incident date/time.

- Vehicle valuation / registration database (external, structured, public):

Check auto_make/model/year against market value to flag inflated claims.

Value, veracity, and variety considerations

- Value: correlate claim/policy features (time-to-claim after binding, claim vs. premium size) with fraud_reported to flag risky claims pre-payout.
- Veracity: claim components may not sum to total_claim_amount, police/valuation records may lag or mismatch reference IDs.
- Variety: structured internal claims/policy data vs. semi-structured external police records vs. structured public vehicle data — each needs different parsing before joining.
- Privacy: age/zip/policy_number are PII, where access should be restricted to claimant PII and police reports.
- Cleaning: de-duplicate claims, reconcile claim sums, handle missing flags, join sources on policy_number/vehicle ID.

Velocity and volume considerations

- Velocity: claims arrive continuously, intake likely real-time/API, enrichment feeds (police, valuation) most likely nightly batch, with spikes after severe weather/multi-vehicle incidents.
- Near real-time: fraud must be flagged before payout, not caught only in an overnight batch.
- Volume: Spans transactional volume (ongoing claim submissions), historical volume (labelled claims needed to train on a minority fraud class), reference/enrichment volume (police and vehicle databases, likely larger than the claims table itself), and per-record footprint (attachments/photos drive storage more than row count).